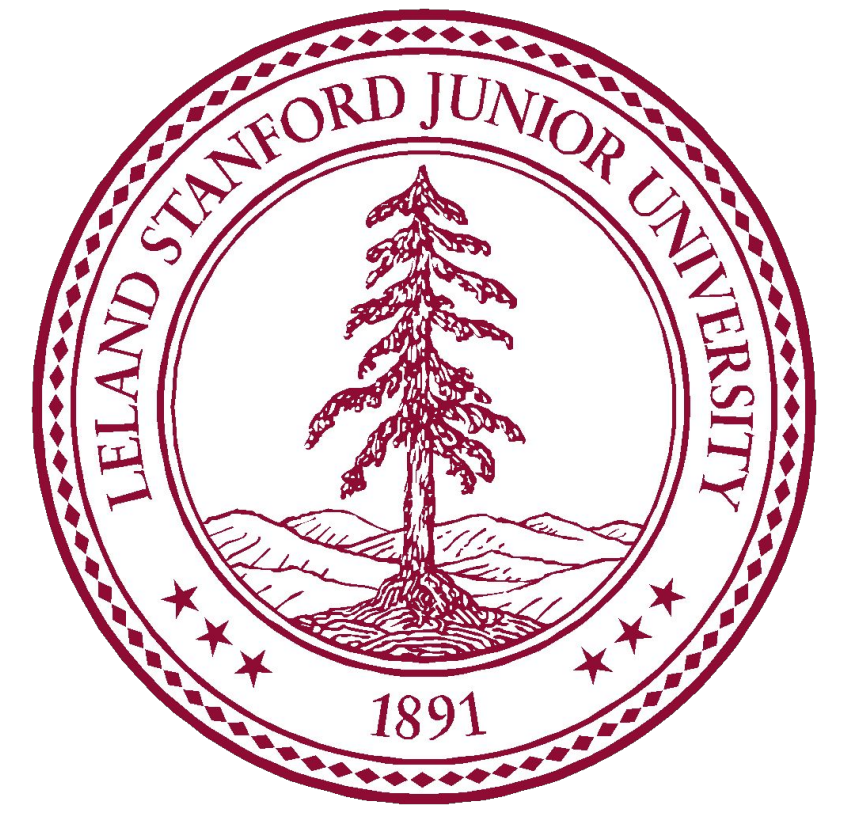


Maturing Multiple Sclerosis Antibodies: How Somatic Hypermutation Generates Pathogenic Autoantibodies Against Myelin Proteins



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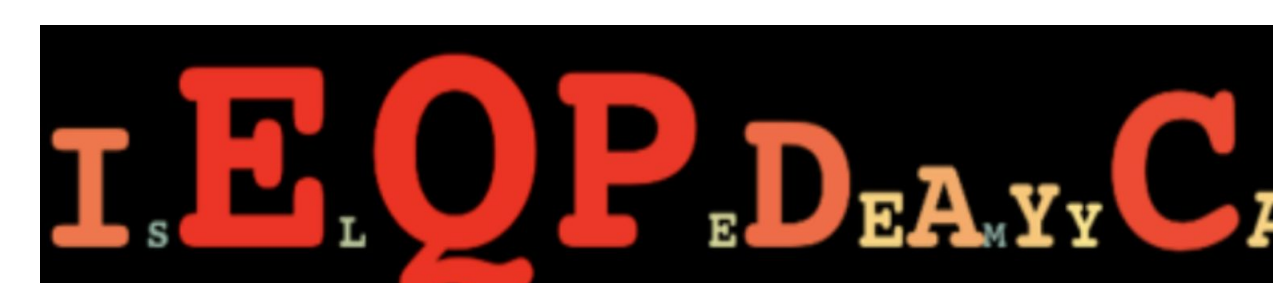
Abstract

Autoreactive B cell activity in the central nervous system (CNS) has been identified as a major hallmark in patients with multiple sclerosis (MS). However, the mechanisms by which these autoreactive B cells develop are unclear. Epstein-Barr virus (EBV), a virus that infects approximately 95% of the population, is considered a prerequisite to developing MS. We hypothesize that anti-EBV antibodies mutate via somatic hypermutation to become autoreactive MS antibodies. Our preliminary data showed that EBV viral proteins, along with broader ExQP motif-containing viral proteins from the herpesvirus family, are recognized by antibodies in the body to eliminate the virus. Eventually, these antibodies develop an affinity for proteolipid protein (PLP). Here, we determine which mutations are responsible for such a switch. To this end, we first used site-directed mutagenesis with PCR to introduce 18 point mutations corresponding to the germline residues. Purified individual plasmids were used to express monoclonal antibodies. Lastly, we performed ELISA assays to determine antibody affinities to PLP and various human peptides with the ExQP motif. Our results showed that multiple mutated antibodies broadly bound to ExQP motifs, supporting the hypothesis that autoimmunity arises via somatic hypermutation of a protein with this motif. Unfortunately, our ELISA results for PLP remained inconclusive. For next steps, we plan to continue a characterization of binding affinities for the remaining 10 mutations. We additionally plan to coat previously established plates with virus-like particles expressing PLP and perform a second ELISA. Successful implementation of this approach will help us understand the development of autoimmunity in MS and the distance necessary for an anti-viral antibody to develop into an antibody against self.

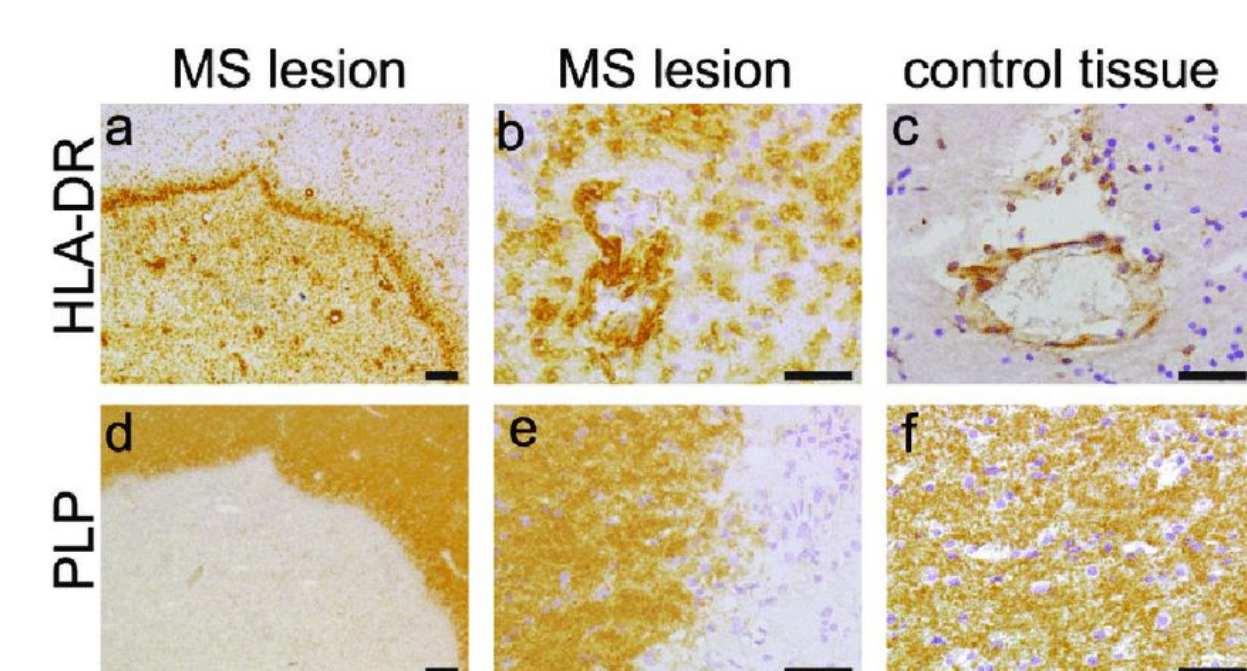
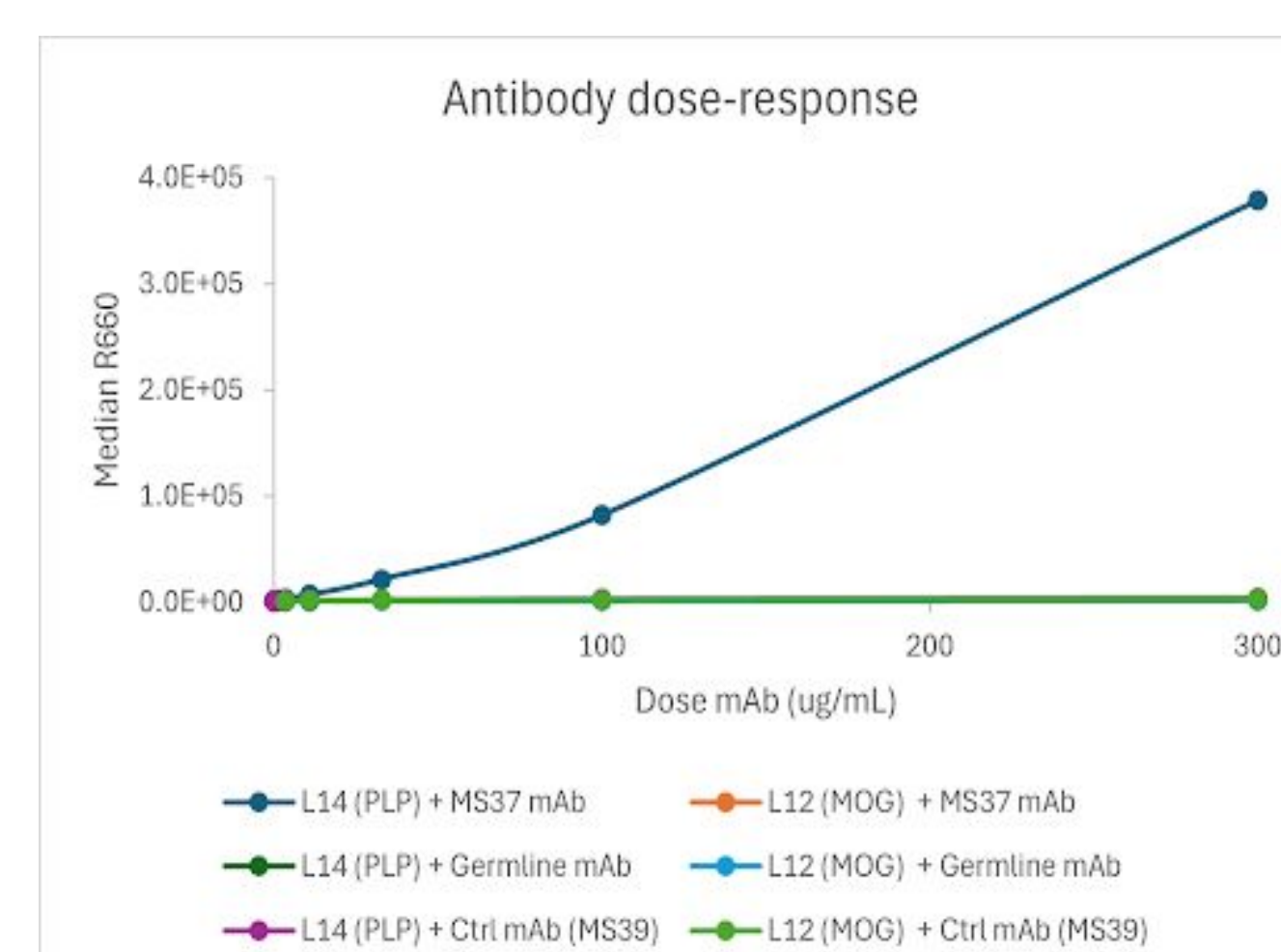
Introduction

- Multiple sclerosis (MS) is an autoimmune disorder in which the central nervous system (CNS) is attacked.
- Proteolipid protein (PLP) comprises over 50% of the CNS, and is targeted by antibodies from MS patients.

- We have found that one MS patient-derived antibody binds to both the ExQP motif from viral EBV proteins and PLP.



- Germline, non-mutated monoclonal antibodies (mAb) have been previously shown to not bind to PLP.



- Epstein-Barr virus (EBV) is one of the most common human viruses, infecting 95% of the population, and is considered as a prerequisite for MS.

Hypothesis

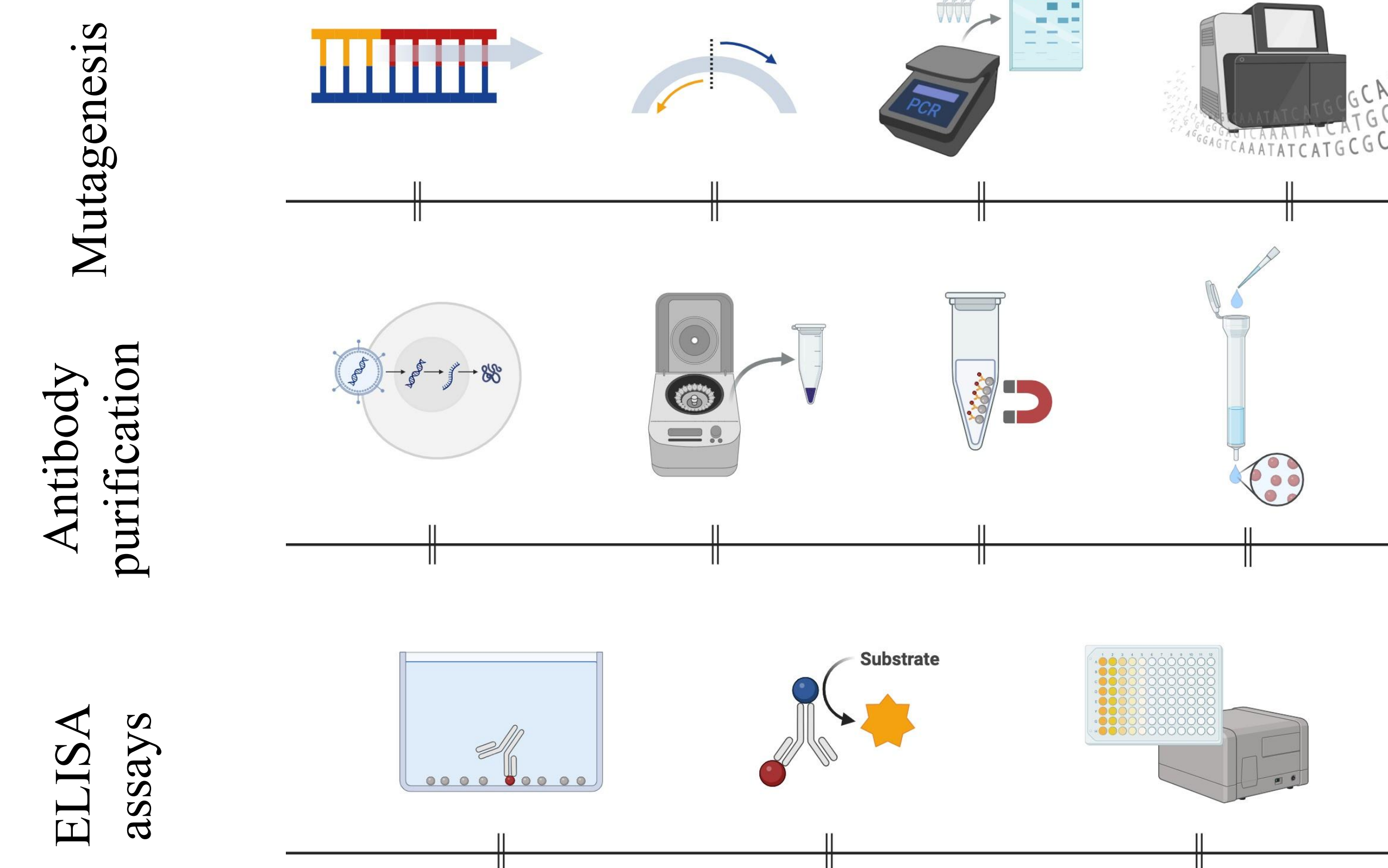
We hypothesized that autoreactive antibodies originate from amino acid point mutations in germline monoclonal antibodies (mAb), following a clonal expansion pathway that shifts their affinity from the ExQP motif to PLP.

Materials and Methods

We performed site-directed mutagenesis via PCR to introduce 18 point mutations that revert the mature autoreactive antibody back to its wild-type (germline) sequence.

Then, we expressed and purified each of these mutated antibodies.

ELISA assays were next performed to determine the antibodies' affinity to various human peptides with the ExQP motif as well as PLP.



Results

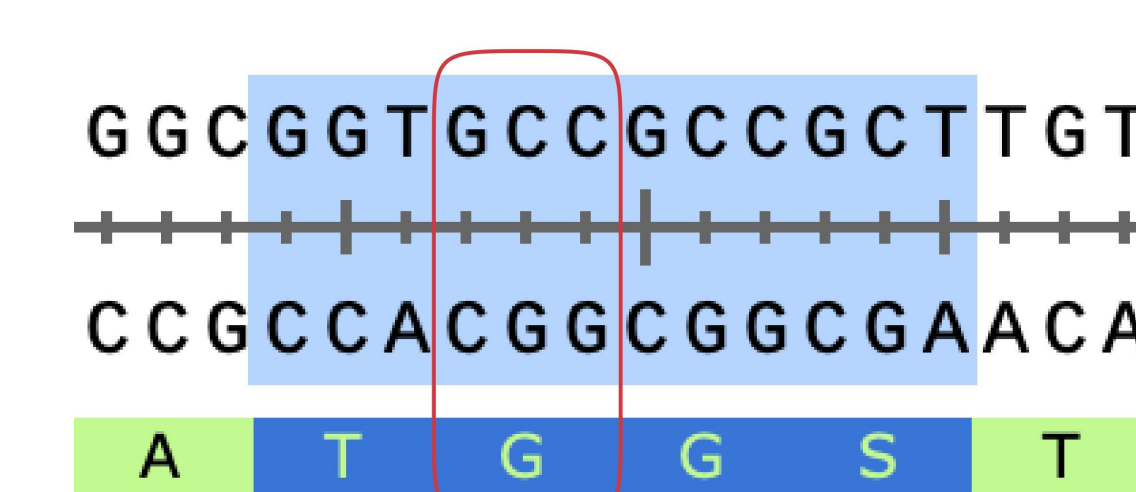


Figure 1. Example sequencing results of mutation H51Y

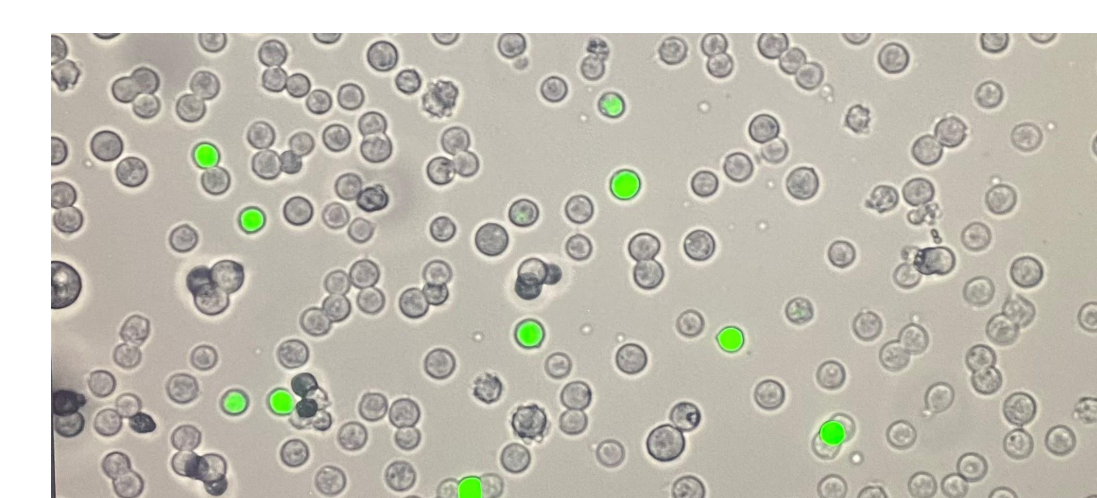


Figure 2. Transfected green fluorescent antibodies

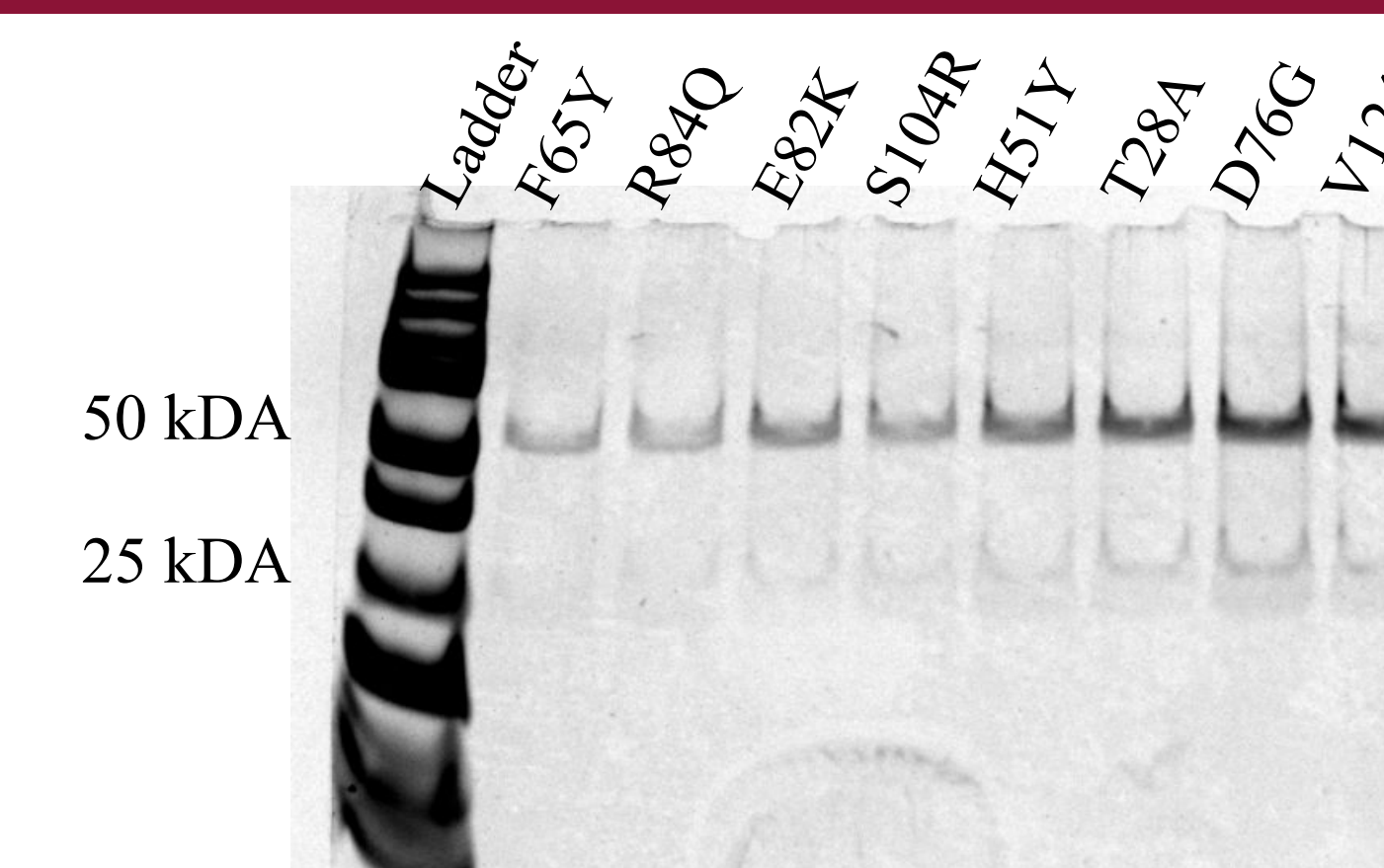
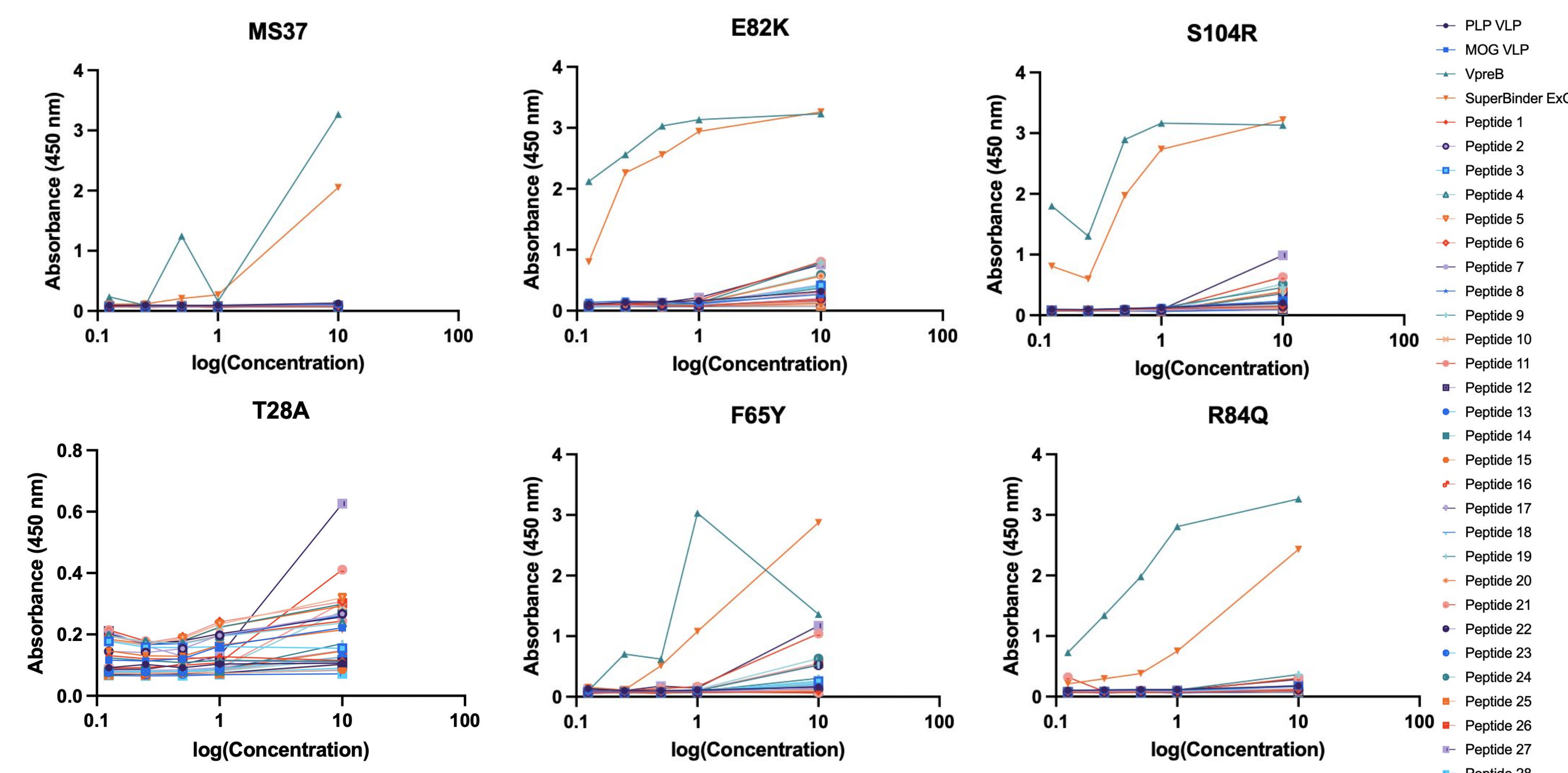


Figure 3. Coomassie gel, with two bands indicating antibodies

Figure 4. ELISA dose response curves, exhibiting broad and increased binding towards ExQP motifs compared to autoreactive antibody



Conclusions

- Multiple mutations that reverted autoreactive antibodies back to germline antibodies exhibited increased binding towards ExQP motifs, supporting the hypothesis that autoimmunity arises via clonal expansion of a protein with this motif
- As predicted by mCSM-AB and using a threshold of $|\Delta\Delta G| > 1$, mutation T51A significantly decreases PLP binding affinity
- PLP results remain inconclusive, as the positive control did not bind

Discussion

Analyzing the remaining 10 mutations will allow for further characterization of MS antibodies' escape mechanisms.

Provided that the computational tool mCSM-AB accurately predicts the relative magnitude and direction of binding affinity change, we hypothesize that the untested mutation T51A contributes to developed PLP affinity.

Under the induced fit model, multiple mutations may create conformational changes. Thus, we are also interested in testing such combinations of mutations.

ELISA plate properties might have impacted VLP binding and antibody measurement. Plate selection will be optimized. In parallel, PLP-binding will be measured with a flow cytometry based assay.

Acknowledgements

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